

A Computational Approach to Retinal-inspired Vision Processing for Edge Detection

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Abstract—Edge detection is a fundamental image processing/computer vision task. Typical algorithms mainly produce fixed gradient or binary edge maps and do not explicitly model retinal mechanisms. To address this gap, this paper introduces a retinal-inspired edge detector by combining adjustable centre-surround receptive fields, ON/OFF bipolar pathways, and threshold modulation to reflect how the retina balances fine-detail acuity and broad structural context. The difference of Gaussian filters with different σ configurations allows the detector to model foveal fine details and parafoveal broader structural representations. Reducing the surround gain below unity further improves performance by preserving centre-dominant edges while maintaining sufficient surround suppression. The ON- and OFF-bipolar pathways enhance local luminance increments and decrements, while threshold modulation of bipolar and amacrine responses refines sensitivity and selectivity under varying visual conditions. The results indicate that the proposed detector provides stronger structural preservation and improved edge quality than classical edge-detection methods. By producing spike-based edge maps, the proposed approach provides a more biologically inspired pre-processing signal for neuromorphic vision tasks.

Index Terms—Retinal-inspired, edge detection, spike encoding, and neuromorphic vision.

I. INTRODUCTION

The retina is the first stage of visual processing, converting light from a visual scene into neural signals through interconnected photoreceptors, horizontal, bipolar, amacrine and ganglion cells, each of which processes visual information locally within its own Receptive Fields (RFs) [1]. Through parallel and asynchronous processing, the retina extracts key features such as edges and motion by detecting spatial and temporal contrast. This feature extraction helps guide visual perception while reducing redundancy in raw visual data, thereby minimising both storage requirements and the computational load for subsequent visual processing. Such compression is evident in the reduction from approximately 126 million photoreceptors to 1.2 million optic nerve fibres, highlighting the role of early visual analysis and filtering in condensing raw visual input into a more compact and efficient representation before it reaches the striate cortex.

Inspired by retinal processing, this paper proposes a neuromorphic detector that uses a centre-surround Difference of Gaussians (DoG) filter to simulate retinal cell responses, extracting and encoding visual stimuli as edge and contrast

information. The filtered signal is then modulated by bipolar excitation and amacrine inhibition to enhance important contrast changes, suppress redundant or noisy activity, and produce a cleaner firing-rate response at the ganglion stage. Spike trains are generated using a Poisson process to reflect the stochastic firing behaviour observed in ganglion outputs [2].

The spike-train representation encodes the retinal model response through firing-rate variations, where stronger contrast produces higher firing rates, weaker contrast produces lower firing rates, and regions with little or no contrast generate minimal spike activity. The resulting firing-rate maps, therefore, provide a qualitative indication of how the proposed Retinal-RF responds to intensity and contrast changes. As opposed to classical edge detectors, which mainly produce gradient-magnitude or binary edge maps [3], [4], the Retinal-RF output provides a graded spike-based response that indicates both the location of edges and the strength of the model's response to those edges and contrast changes. These spike trains provide rich and dynamic input signals for advanced spiking neural networks (SNNs), where spike-based computation can further refine edge-detection results in an energy-efficient manner which can serve as a fundamental preprocessing stage for downstream vision tasks such as object recognition, scene analysis, and autonomous navigation. The main contributions of this paper are the following:

- A retinal-inspired neuromorphic edge detector is proposed by combining adjustable centre-surround DoG filtering, bipolar excitation, and amacrine inhibition to produce selective ganglion-cell firing-rate representations.
- The firing-rate maps encode intensity/contrast changes as spike activity making Retinal-RF suitable for SNN-based vision and downstream neuromorphic tasks.

II. RELATED WORK

Edge Detectors: An edge is an abrupt local change in image intensity or colour, often detected using derivative-based methods. Existing edge detectors are broadly classified into first-derivative gradient operators and second-derivative techniques [5]. Gradient-based methods include the Robert's operator which provides a simple and fast estimate of the 2D spatial gradient in an image [6]. Prewitt and Sobel operators have combined into a hybrid kernel whose weights

are optimised using an ANN to improve edge precision and image-segmentation performance [7]. Canny is a multi-stage edge detector designed to identify sharp intensity changes and produce well-localised image boundaries [8]. The Marr-Hildreth algorithm applies LoG filtering and detects edges at zero-crossings where the response changes sign [4]. These operators are computational simple and efficient but can struggle with noise, weak edges, and complex scenes [3]. Gabor filters are used for edge detection and texture analysis by capturing spatial frequency and orientation, and are inspired by visual-cortex simple-cell receptive fields [9], rather than the retina.

Retinal processing: Photoreceptors convert light into electrical signals, while horizontal cells provide lateral inhibitory feedback that enhances local contrast and sharpens edge responses by suppressing activity in uniform regions and emphasising rapid brightness changes [1]. This local modulation varies across the retina, with the foveal region supporting fine spatial detail and precise edge information. The parafoveal region balances local detail with broader contextual information, and the peripheral retina integrates wider visual regions, making it more sensitive to global changes and motion. Thus, retinal processing adapts across foveal, parafoveal, and peripheral regions to support both detailed visual analysis and efficient scene-level perception.

The modulated signals are transmitted to bipolar cells, where ON and OFF types selectively respond to increments and decrements in light intensity, respectively. The bipolar cells release glutamate onto amacrine and ganglion cells, helping determine whether downstream cells respond to light increments or decrements. Amacrine cells then modulate this signalling through inhibitory interactions, shaping the final ganglion-cell output [10]. They refine spatio-temporal contrast through modulation of ganglion-cell responses, helping the retina detect where and when stimuli change, including rapid intensity changes, motion timing, and movement direction.

Ganglion cells generate action potentials to encode visual information for long-distance transmission through the optic nerve. They represent spatial and temporal features such as contrast, edges, and motion through RFs shaped by upstream retinal circuits, including amacrine, bipolar, and horizontal cells. Depending on their RF organisation, ganglion cells can exhibit ON-centre or OFF-centre responses, where ON-centre cells respond strongly to light increments in the centre of the RF, while OFF-centre cells respond strongly to light decrements. In summary, retinal cells pre-process visual signals into a compact representation of contrast, edges, and motion, which is relayed via the optic nerve and lateral geniculate nucleus to the striate cortex (V1) for further feature extraction, including orientation-selective edge processing and more refined spatial analysis. This locally organised interaction between cells enables efficient visual processing and illustrates how biological vision reduces redundancy before higher-level perception of shape, depth, and movement.

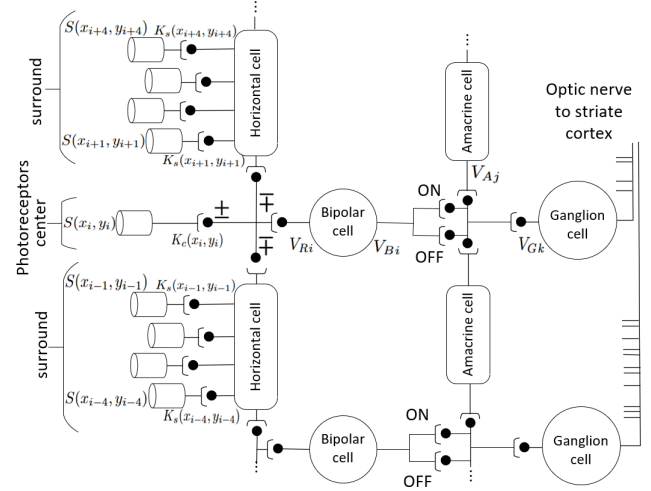


Fig. 1: Structure of the proposed retina model, where photoreceptors convert light into V_{Ri} after horizontal-cell regulation. This signal is processed by i^{th} bipolar-cell to produce V_{Bi} , further modulated by signal V_{Aj} from j^{th} amacrine to generate V_{Gk} for k^{th} ganglion-cell to produce spike trains.

III. RETINA NETWORK MODEL

The retina is organised in five neuronal types such as photoreceptors, horizontal, bipolar, amacrine and ganglion cells [1]. As shown in Fig. 1, these cells form two main connectivity structures: a feed-forward pathway from photoreceptors to bipolar and ganglion cells, which converts light into neural signals and ultimately spike-based outputs; and lateral pathways in which horizontal cells shape centre-surround RF structures while amacrine cells refine spatial and temporal contrast by modulating ganglion-cell activity. This paper aims to develop a retinal model with adjustable horizontal-, bipolar-, and amacrine-cell parameters to examine how inter-neuronal interactions shape ganglion-cell firing dynamics in response to visual scenes. By analysing these parameter-driven interactions, the proposed approach provides insight into how retinal circuits shape visual output and supports a deeper understanding of early visual processing.

Visual scenes, which are spatially inhomogeneous and non-stationary, can be described by a spatiotemporal function $S(x, y, t)$ where contrast levels are confined to a single channel within the range $[0, 1]$ here, and the function S depends on both spatial coordinates (x, y) and time t . The receptor-driven signal (V_{Ri}) that activates i^{th} bipolar-cell can be denoted by Eq. (1) to define how the spatial stimulus at time t is integrated and weighted by the bipolar's RF kernel, $K_{Bi} = f(K_c, K_s)$.

$$V_{Ri}(t) = [K_{Bi} * S](t), \quad (1)$$

The i^{th} bipolar-cell is located at position (x_i, y_i) in the visual field, which is also the center of its RF. The RFs of bipolar cells in both the foveal and parafoveal regions can be modelled by using a center-surround DoG structure. In the fovea, RFs are smaller and more spatially localised, while in the parafoveal, RFs are broader and less sharply tuned, reflecting

the trade-off between resolution and coverage across the visual field. The temporal RF can be modelled with a Gamma kernel [11], which produces a uniphasic response with parameter-controlled delay and sharpness, ideal for transient ON or OFF dynamics. Alternatively, a difference of two cascaded Gamma models [12] can be employed to generate a biphasic response capturing both excitatory/ON and inhibitory/OFF dynamics, which is more biologically accurate for many sensory neurons including retina. This paper establishes the spatial correctness of the model as a baseline, and temporal validation will be considered in future work.

The excitatory signals from bipolar cells are received and modulated by amacrine cells through their lateral connectivity. Amacrine cells, which form inhibitory synapses onto both bipolar cells and ganglion cells, enable feedback and lateral inhibition within the inner retina. This modulation refines the spatial (and temporal) properties of visual signals to support functions such as contrast enhancement, the generation of transient responses, and motion detection. By regulating the flow of information from bipolar to ganglion cells, amacrine cells introduce an additional layer of processing that enhances the complexity and precision of retinal output. This interaction can be simplified through Eq. (2) to Eq. (5).

$$V_{B_i}(t) = V_{R_i}(t) + \sum_{j=1}^{N_A} W_{A_j B_i} \mathcal{N}_A(V_A). \quad (2)$$

The term $\sum_{j=1}^{N_A} W_{A_j B_i} \mathcal{N}_A(V_A)$ represents the sum of inputs from all the connected amacrine cells ($j = 1, \dots, N_A$), each weighted by its synaptic strength $W_{A_j B_i} \leq 0$ (only inhibitory amacrine cells are considered here), and passed through the threshold-linear function ($\mathcal{N}_A$) as in Eq. (3):

$$\mathcal{N}_A(V_A) = \begin{cases} V_A - A_0, & \text{if } V_A > A_0; \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

where threshold A_0 is identical for all amacrine cells. Amacrine and bipolar cell relationship can be described as:

$$V_{A_j}(t) = \sum_{i=1}^{N_B} W_{B_i A_j} \mathcal{N}_B(V_B), \quad (4)$$

where $W_{B_i A_j} \geq 0$ and $\mathcal{N}_B$ has the same function as (3) with a threshold B_0 . Ganglion cells receive excitatory synaptic inputs from bipolar cells with weights $W_{B_i G_k} \geq 0$, and inhibitory inputs from amacrine cells with weights $W_{A_j G_k} \leq 0$. Their overall response can be expressed by Eq. (5).

$$V_{G_k}(t) = \sum_{j=1}^{N_A} W_{A_j G_k} \mathcal{N}_A(V_A) + \sum_{i=1}^{N_B} W_{B_i G_k} \mathcal{N}_B(V_B). \quad (5)$$

IV. RESULTS AND DISCUSSION

The input image (Fig. 2a) is processed using the proposed DoG filter (K_{B_i}) with all the weights are set at constant ($W_{A_j B_i} = W_{A_j G_k} = -1$ and $W_{B_i A_j} = W_{B_i G_k} = 1$), to simulate bipolar RFs. Different σ values are applied to approximate the RFs of foveal ($\sigma_c = 1$, $\sigma_s = 3$) and

parafoveal ($\sigma_c = 3$, $\sigma_s = 5$) regions, resulting in distinct visual representations as shown in Fig. 2b & 2c.

In the foveal condition, small σ values are applied to create narrow RFs that emphasise fine spatial detail, as shown in Fig. 2b. The V_B representation corresponds to the bipolar response, which captures the local structure of the input but still contains some surrounding activity. After modulation at the ganglion stage, the V_G representation becomes sharper, with stronger foreground responses and reduced background activity. The firing-rate map is generated by converting ganglion response strength into spike rates using a Poisson process. It shows higher activity in regions with light increments and strong positive contrast, *i.e.*, digit strokes, while uniform background regions generate little or no response. The spike map and ganglion response plots further indicate that spike activity is mainly concentrated around strong contrast regions. This encoding enables precise discrimination of fine structures, essential for tasks such as reading, recognition, and subtle shape detection. Furthermore, the approach reduces redundancy in uniform regions, optimising information transfer for high-resolution analysis. This property reflects the biological role of the fovea, where densely packed cone photoreceptors and narrowly tuned RFs support high-acuity vision [1].

In parafoveal condition, larger σ values produce smoother responses (see Fig. 2c). Compared with the foveal condition, the V_B and V_G representations appear broader and less sharply localised, indicating that fine spatial details are suppressed while luminance transitions are preserved. The firing-rate map and spike activity are spread over wider regions of the digit strokes, producing a coarser representation that preserves the main structure while reducing sensitivity to details, noise, and high-frequency variations. This makes the parafoveal representation suitable for detecting broader shapes, object outlines, and contrast boundaries, where contextual awareness is more important than fine-detail discrimination. Bipolar RFs generate separate ON- and OFF-bipolar maps, where ON pathways emphasise local luminance increments and OFF pathways emphasise luminance decrements relative to the surround. As shown in Fig. 2b and Fig. 2d, these complementary maps capture both positive and negative contrast boundaries, enhancing edge structure while suppressing uniform regions.

Figure 3 shows how adjusting the cell thresholds ($0 < \text{thresholds} \leq 1$ and $A_0 \geq B_0$) affects retinal ganglion responses, thereby regulating the generated spikes. When the bipolar threshold (B_0) is reduced while the amacrine threshold remains constant ($A_0 = 1$), the firing-rate maps become progressively more responsive to input signals (Fig. 3a). This indicates that lowering B_0 increases retinal sensitivity by allowing weaker inputs to activate bipolar cells and generate denser ganglion spike responses, although this may also reduce selectivity and increase noise. On contrary, lowering the amacrine threshold A_0 produces stronger modulation, shaping the spatial spread of ganglion responses by enhancing edge continuity while potentially blurring fine details. These findings align with the known roles of bipolar and amacrine cells in the retina (see Section II), where bipolar cells pro-

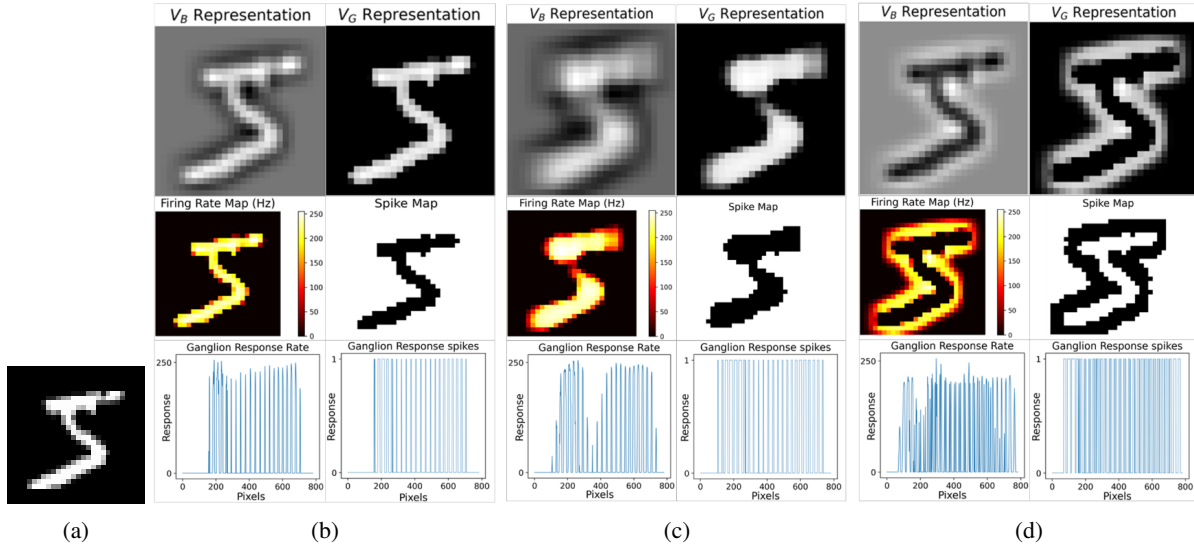


Fig. 2: Representations produced by different configurations of our retinal-inspired detector: (a) input; (b) ON-bipolar foveal, (c) ON-bipolar parafoveal, and (d) OFF-bipolar foveal outputs. Each configuration shows the V_B and V_G representations, firing rate maps, spike maps and ganglion response maps, demonstrating how foveal and parafoveal settings preserve fine detail and capture broader structural features, while ON- and OFF-bipolar pathways highlight edge transitions.

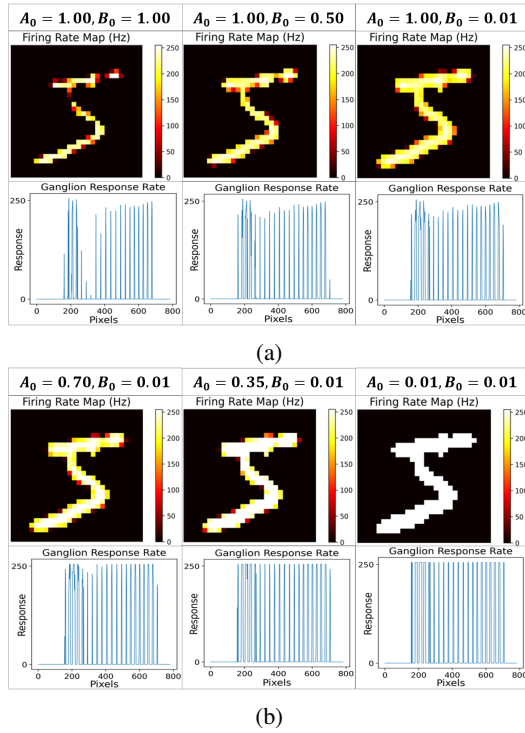


Fig. 3: Effects of reducing (a) bipolar (B_0) and (b) amacrine (A_0) thresholds on retinal ganglion responses. From left to right, PSNR(dB): 12.79, 16.19, 17.88, 17.46, 16.18, 14.38. SSIM: 0.5053, 0.7673, 0.8446, 0.8401, 0.8151, 0.7700.

vide the excitatory drive to ganglion cells, while amacrine cells introduce inhibitory modulation that shapes ganglion's responses. Adjusting these thresholds mimics how the retina balances sensitivity vs. selectivity depending on environmental conditions such as low light versus high contrast scenes.

The proposed detector is evaluated against classical edge detectors implemented using OpenCV, with the full quantitative Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) results presented in Table I. The results show that the proposed Retinal-RF method consistently outperformed the classical methods across all ten test images, achieving the highest PSNR and SSIM values for each image as well as the strongest average performance overall. Specifically, Retinal-RF achieved a 32.57% improvement in PSNR compared with Sobel and a SSIM value approximately 2.51 times higher than Prewitt. This suggests that Retinal-RF captures clearer object boundaries while preserving object shapes, contours, and structural organisation more effectively than the classical methods.

For the Retinal-RF methods, the foveal configuration was implemented with $A_0 = 0.99$, $B_0 = 0.01$, centre gain $K_c = 1$ and reduced surround gains $K_s = 0.99$ and $K_s = 0.98$. Setting $K_s < 1$ introduces a mild effect of bio-inspired surround-suppression behaviour [13], where the surround provides inhibitory modulation without fully cancelling the centre response. Among the tested settings, $K_s = 0.98$ outperformed $K_s = 0.99$, suggesting that a slightly reduced surround gain helps preserve centre-dominant edge information while maintaining sufficient surround suppression to reduce responses in uniform regions and enhance local contrast boundaries. In contrast, conventional computer-vision DoG filtering often set the centre and surround gains equally to one to perform band-pass filtering and highlight rapid intensity changes [14], [15].

Table II presents the visual comparison between classical edge detectors and the proposed Retinal-RF method. For clearer visualisation, the edge maps were intensity-inverted using $255 - I$. Roberts, Prewitt, and Sobel are first-derivative operators that produce gradient-based edge responses. How-

TABLE I: Full PSNR and SSIM results for the ten test images.

Methods	Camera	Coins	Moon	Page	Text	Retina	Astronaut	Checkerboard	Clock	Cat	Average
PSNR (dB)											
Roberts	5.2935	8.7358	7.2918	3.6159	6.4461	8.7559	6.1066	3.5273	4.8680	7.1552	6.1796
Sobel	5.7078	9.9165	7.7788	4.0951	7.1768	9.2707	6.7692	3.8686	5.0398	8.5107	6.8134
Prewitt	5.5767	8.8118	7.7796	2.9293	7.0625	9.1855	6.1547	0.5354	5.0441	8.3408	6.1420
Canny (100, 200)	4.7703	7.5340	7.0425	3.0392	5.5391	8.5085	5.3014	3.2012	4.7452	6.2048	5.5886
LoG (ksize=3)	5.4532	9.2266	7.4628	3.7567	6.6981	8.8876	6.2980	3.8703	4.9180	7.3555	6.3927
Retinal-RF ($K_s = 0.99$)	6.4485	10.0362	8.3138	5.3125	7.8835	10.0100	7.6486	6.5378	6.1807	8.3461	7.6718
Retinal-RF ($K_s = 0.98$)	7.6775	10.7814	9.7564	6.6069	9.0402	11.7817	8.9052	8.1435	8.2213	9.4080	9.0322
SSIM											
Roberts	0.0969	0.1141	0.0234	-0.1100	-0.0328	0.2273	0.1978	0.2813	0.0171	0.0230	0.0838
Sobel	0.0927	0.1256	0.0747	-0.0206	0.0040	0.2772	0.1426	0.1905	0.0379	0.0535	0.0978
Prewitt	0.2023	0.2293	0.0848	-0.0137	0.0171	0.2786	0.2464	0.1794	0.0462	0.1043	0.1375
Canny (100, 200)	0.0151	0.0230	0.0004	-0.0212	-0.0234	0.2167	0.1176	0.2807	0.0002	-0.0013	0.0608
LoG (ksize= 3)	0.0962	0.1176	0.0457	-0.1452	-0.0571	0.2097	0.1548	0.1638	0.0222	0.0319	0.0640
Retinal-RF ($K_s = 0.99$)	0.2367	0.2547	0.1281	0.2532	0.1822	0.3266	0.3463	0.5877	0.1932	0.1919	0.2701
Retinal-RF ($K_s = 0.98$)	0.3185	0.2905	0.2208	0.3187	0.2189	0.3993	0.3977	0.6414	0.3961	0.2564	0.3458

ever, weak or low-contrast boundaries may appear faint or fragmented, as seen in the Coins image, where the main outlines are visible but subtle surface details are less preserved. In Table I, Sobel achieved the highest PSNR, indicating lower pixel-level distortion, while Prewitt achieved the highest SSIM, suggesting stronger structural similarity. However, Sobel performs less effectively on the Moon and Clock images due to smoother regions and fewer strong boundary transitions, while Prewitt is weaker on the Checkerboard image due to the strong difference between its edge map and the original square pattern. Overall, Sobel produces clearer object outlines than Roberts and Prewitt, but its edge responses are often thicker and contain lighter interruptions along the edge bands.

Canny uses smoothing, non-maximum suppression, and hysteresis thresholding to reduce noise and produce thin, well-localised edges, but its output is sensitive to threshold selection. As shown in the Retina image, the circular boundary is strongly detected, while weaker internal details are suppressed by the selected low and high thresholds of 100 and 200. LoG detects zero-crossings after Gaussian smoothing, producing thin and localised contours; however, its second-derivative nature can create double-edge responses around strong boundaries, as seen in the Checkerboard image. In lower-contrast images such as Cat or Coins, weaker details, *i.e.*, inner coin and whisker textures appear fragmented. On the other hand, the proposed Retinal-RF method preserves clearer object boundaries and structural details across different image types, while suppressing large uniform regions. This is evident in the Checkerboard image, where Retinal-RF retains the repeated square arrangement, whereas the classical methods mainly produce intersecting line responses. This visual behaviour supports the quantitative PSNR and SSIM results, where Retinal-RF achieved stronger structural preservation than classical methods.

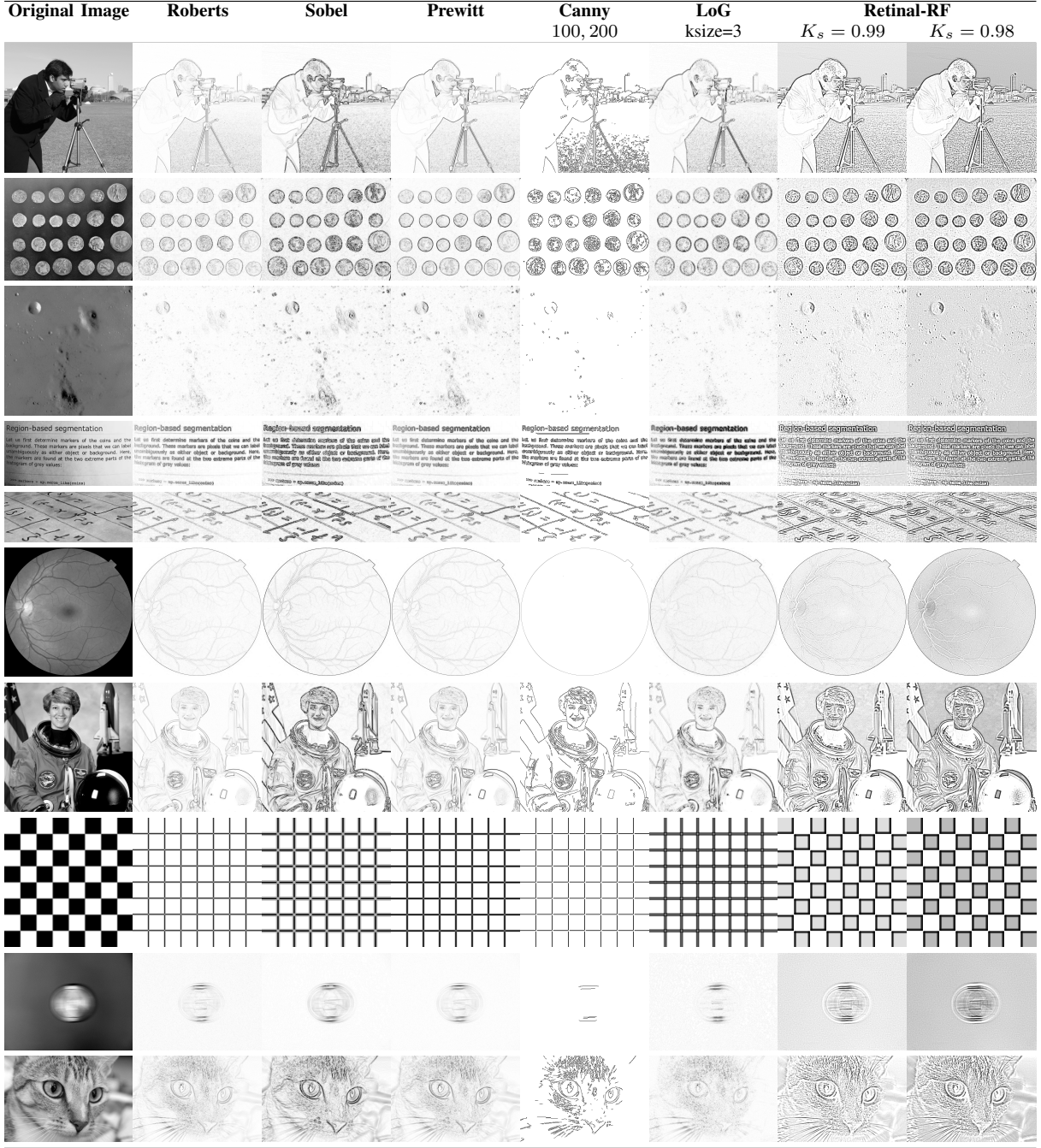
V. CONCLUSION

This paper presented a retinal-inspired neuromorphic edge detector that combines centre-surround RFs, ON/OFF bipolar pathways, threshold modulation, and spike-based encoding. The proposed Retinal-RF method achieved stronger structural preservation and improved visual edge quality compared with classical operators, while maintaining sensitivity to varying contrast conditions. The firing-rate maps also provided a qualitative indication of dynamic range handling by showing how stronger image features produce higher spike activity. Overall, the results suggest that retinal-inspired pre-processing can enhance feature extraction for downstream vision tasks such as object recognition, scene analysis, image segmentation, and spiking neural network training, helping bridge classical computer vision with biologically inspired visual encoding. Future work will explore automatic adaptation of A_0 , B_0 and K_s , enabling the model to dynamically tune sensitivity, selectivity, and surround suppression based on input-scene properties such as entropy, contrast and illumination.

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TABLE II: Visual comparison results. Below displayed edge maps were inverted using $255 - I$ to improve visual clarity and make edge structures easier to observe. This inversion was not applied to the quantitative metric calculations.



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